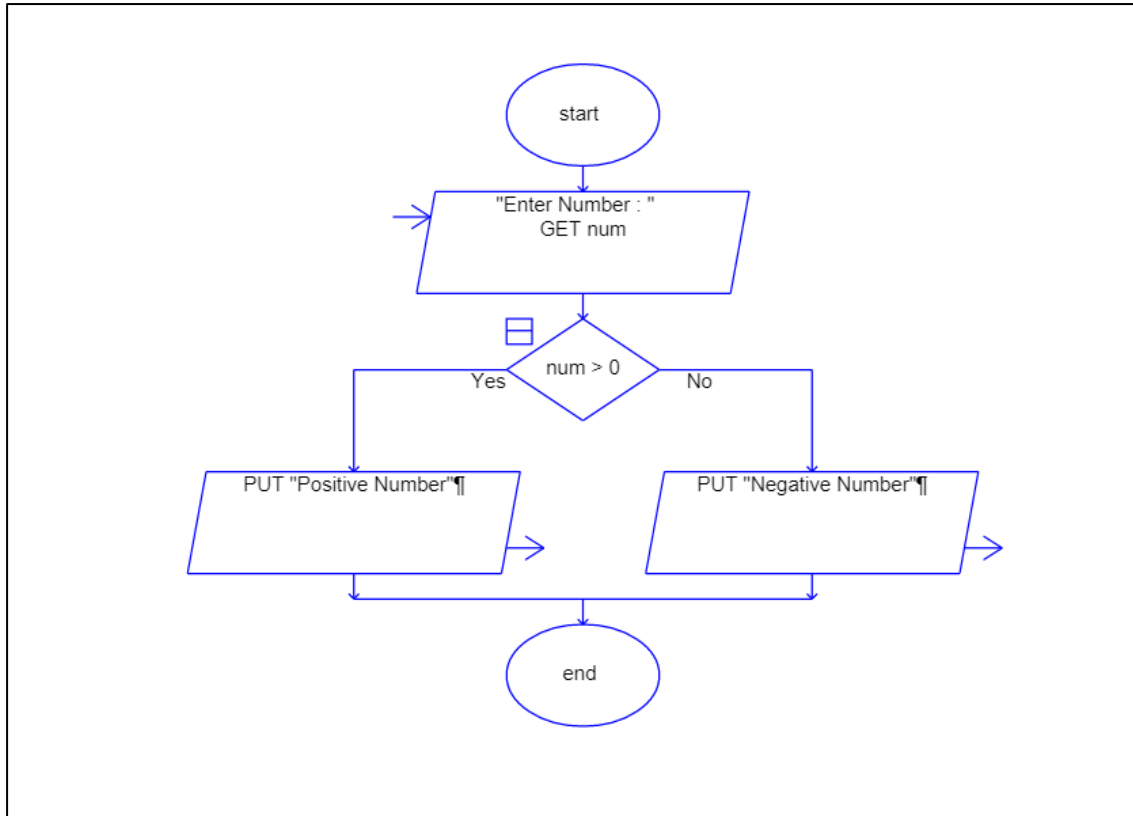


CDAC Mumbai

Lab Assignment: Flowchart and Java Programming

1. Check Positive Number:

- Task: Create a flowchart to check whether a number is positive.



- Next Step: Write a Java program that checks if a predefined number is positive using an if-else statement and prints the appropriate message.

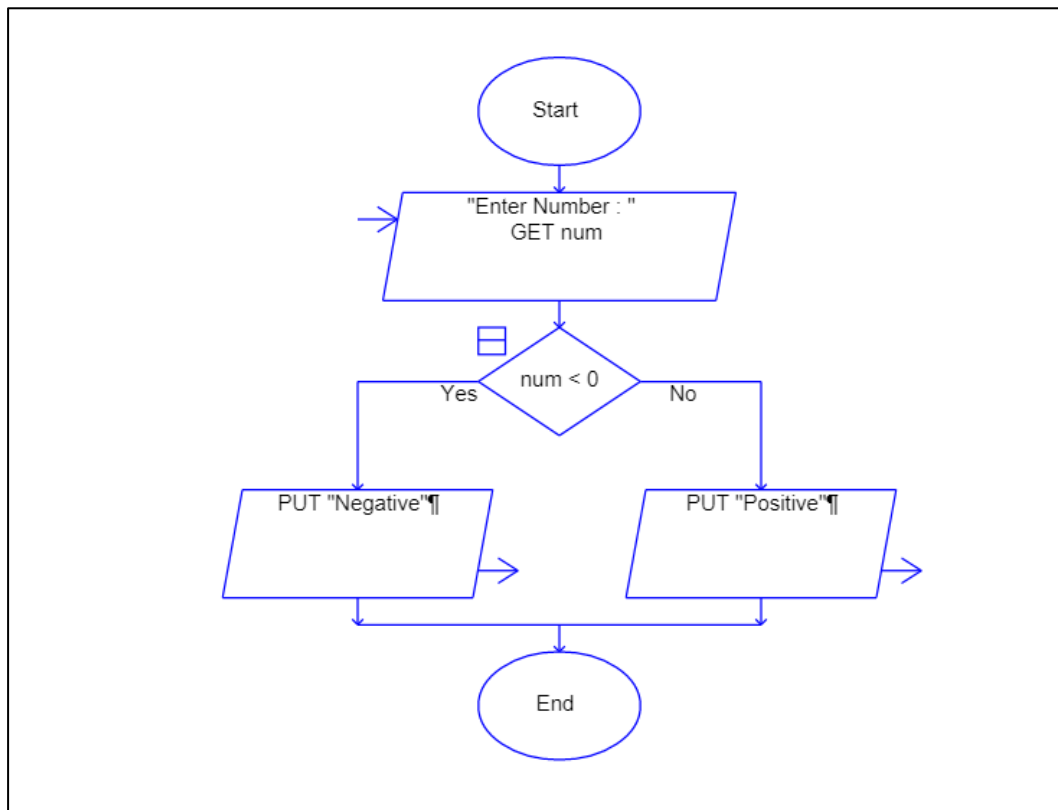
```
1 public class positivenumber{
2     Run | Debug
3     public static void main(String args[]){
4         int num = 15;
5
6         if(num > 0){
7             System.out.println(x: "Positive Number");
8         }
9         else{
10            System.out.println(x: "Negative Number");
11        }
12    }
13 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS powershell

```
● PS C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 1\Assignment> java positivenumber.java
Positive Number
○ PS C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 1\Assignment> 
```

2. Check Negative Number:

- Task: Create a flowchart to check whether a number is negative.



- Next Step: Write a Java program that checks if a predefined number is negative using an if-else statement and displays the result.

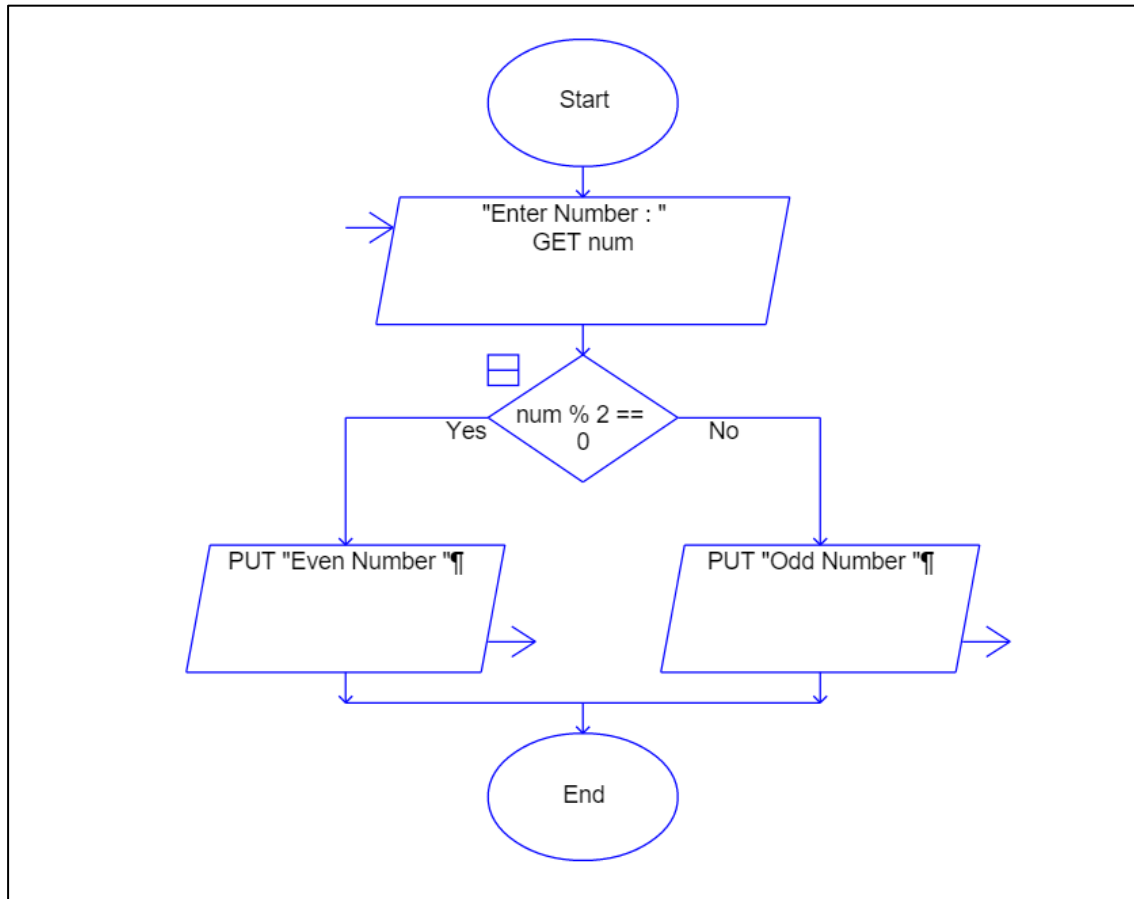
```
negativenumber.java > negativenumber > main(String[])
1 public class negativenumber {
2     public static void main(String args[]){
3         int num = -15;
4
5         if(num < 0){
6             System.out.println(x: "Negative Number");
7         }
8         else{
9             System.out.println(x: "Positive Number");
10        }
11    }
12 }
13
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS powershell

```
PS C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 1\Assignment> java negativenumber.java
Negative Number
PS C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 1\Assignment>
```

3. Check Odd or Even Number:

- Task: Create a flowchart to determine whether a number is odd or even.



- Next Step: Write a Java program that checks if a predefined number is odd or even. Use an if-else statement and the modulus operator (%) to determine whether the number is divisible by 2 or not.

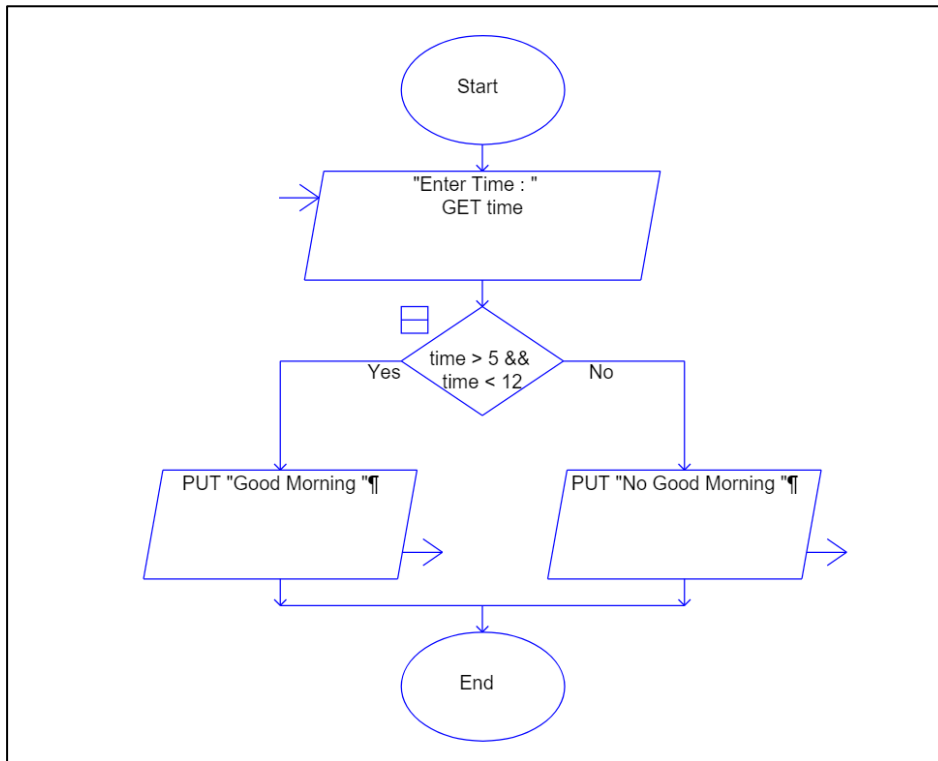
```
J evenodd.java > evenodd
1  public class evenodd {
2      Run | Debug
3      public static void main(String[] args) {
4          int num = 20;
5
6          if(num % 2 == 0){
7              System.out.println(x: "Even Number ");
8          }
9          else{
10             System.out.println(x: "Odd Number ");
11         }
12     }
13 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
● PS C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 1\Assignment> java evenodd.java
Even Number
○ PS C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 1\Assignment>
```

4. Display Good Morning Message Based on Time:

- Task: Create a flowchart to display a "Good Morning" message based on a given time.



- Next Step: Write a Java program that displays a "Good Morning" message if the predefined time is between 5 AM and 12 PM. Use an if statement to implement the logic.

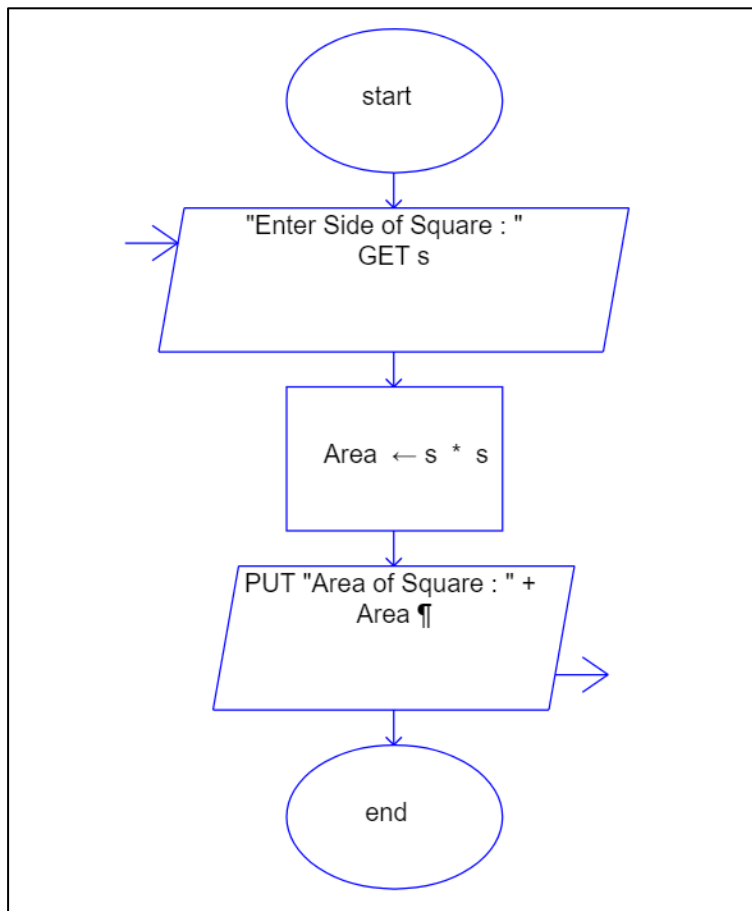
```
gm.java > gm > main(String[])
1  import java.util.*;
2
3  public class gm {
4      public static void main(String[] args) {
5          Scanner sc = new Scanner(System.in);
6          System.out.print(s: "Enter time : ");
7          int time = sc.nextInt();
8
9          if(time > 5 && time < 12){
10             System.out.println(x: "Good Morning ");
11         }
12     }
13 }
14
```

PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logi Building 1\Assignment> java gm.java
Enter time : 7
Good Morning
PS C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logi Building 1\Assignment>
```

5. Print Area of a Square:

- Task: Create a flowchart to calculate and print the area of a square.



- Next Step: Write a Java program that calculates the area of a square using the formula $\text{area} = \text{side} * \text{side}$. Use a predefined side length.

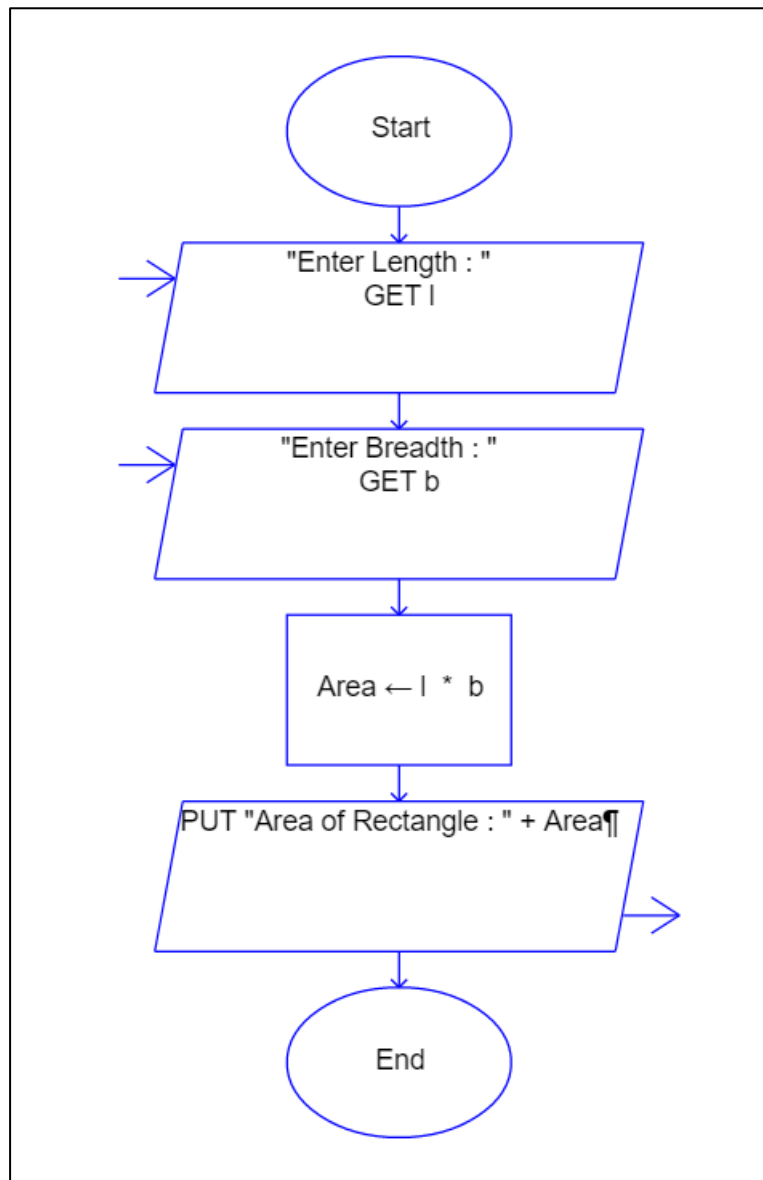
```
J square.java > ...
1  import java.util.*;
2
3  public class square {
4      Run | Debug
5      public static void main(String[] args) {
6          Scanner sc = new Scanner(System.in);
7          System.out.print(s: "Enter side : ");
8          int side = sc.nextInt();
9
10         int area = side * side;
11         System.out.print("Area of Square : " + area);
12     }
13 }
```

PROBLEMS 2 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
● PS C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 1\Assignment> java square.java
Enter side : 5
Area of Square : 25
○ PS C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 1\Assignment> 
```

6. Print Area of a Rectangle:

- Task: Create a flowchart to calculate and print the area of a rectangle.



- Next Step: Write a Java program that calculates the area of a rectangle using the formula $\text{area} = \text{length} * \text{width}$. Use predefined values for length and width.

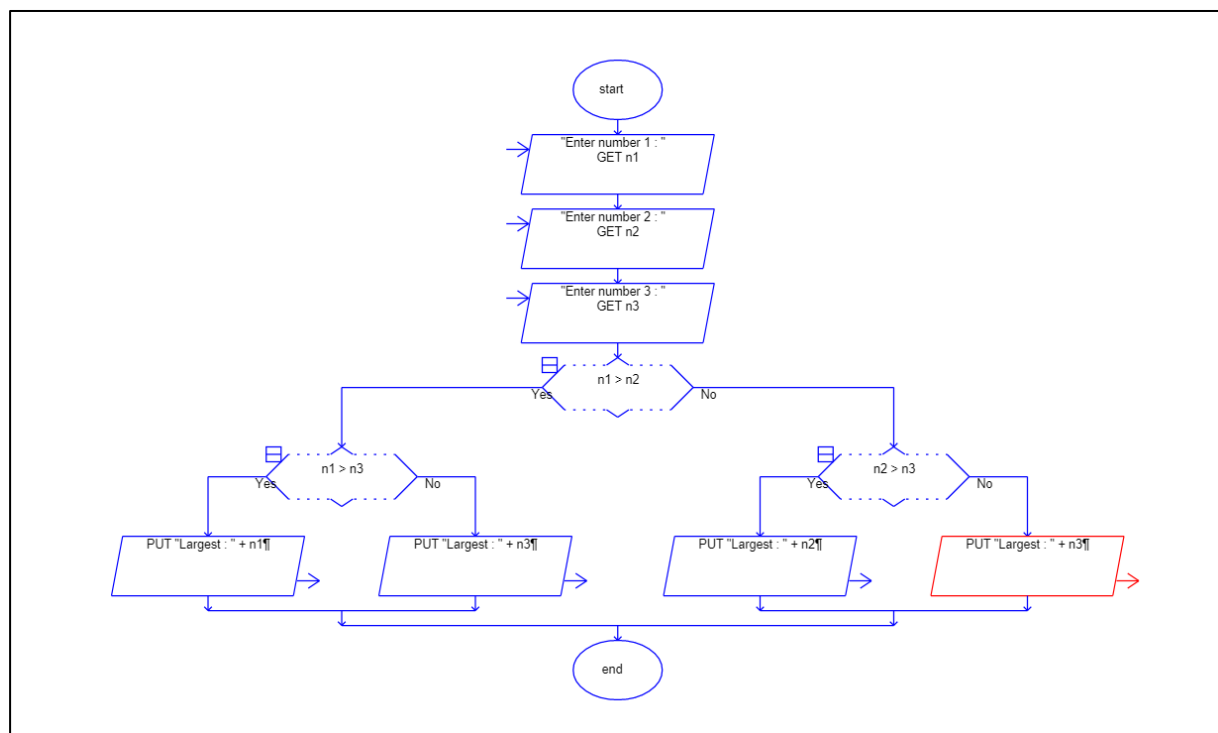
```
J rectangle.java >  rectangle
1  import java.util.*;
2
3  public class rectangle{
    Run | Debug
4      public static void main(String args[]){
5          Scanner sc = new Scanner(System.in);
6          System.out.print(s: "Enter Length : ");
7          int length = sc.nextInt();
8          System.out.print(s: "Enter Breadth : ");
9          int breadth = sc.nextInt();
10
11         System.out.println("Area of Rectangle : " + (length * breadth));
12     }
13 }
```

Problems 3 Output Debug Console **Terminal** Ports

- PS C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 1\Assignment> java rectangle.java
Enter Length : 3
Enter Breadth : 2
Area of Rectangle : 6
- PS C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 1\Assignment> |

7. Find the Largest of Three Numbers:

- Task: Create a flowchart to find the largest of three numbers.



- Next Step: Write a Java program that finds and prints the largest of three predefined numbers using if-else statements.

J Largestof3.java > ...

```

1  import java.util.*;
2
3  public class Largestof3 {
    Run | Debug
4      public static void main(String[] args) {
5          Scanner sc = new Scanner(System.in);
6          System.out.print(s: "Enter number 1 : ");
7          int n1 = sc.nextInt();
8          System.out.print(s: "Enter number 2 : ");
9          int n2 = sc.nextInt();
10         System.out.print(s: "Enter number 3 : ");
11         int n3 = sc.nextInt();
12
13         if(n1 > n2){
14             if(n1 > n3){
15                 System.out.println("Largest : " + n1);
16             }
17             else{
18                 System.out.println("Largest : " + n3);
19             }
20         }

```

```

19         }
20     }
21     else{
22         if(n2 > n3){
23             System.out.println("Largest : " + n2);
24         }
25         else{
26             System.out.println("Largest : " + n3);
27         }
28     }
29 }
30 }
31

```

Problems 4 Output Debug Console Terminal Ports

```

● PS C:\Users\vnaik\Desktop\CDAC\CDAC Course\Logic Building 1\Assignment> java Largestof3.java
Enter number 1 : 30
Enter number 2 : 10
Enter number 3 : 50
Largest : 50

```